

and can form the X-ray spectrum.

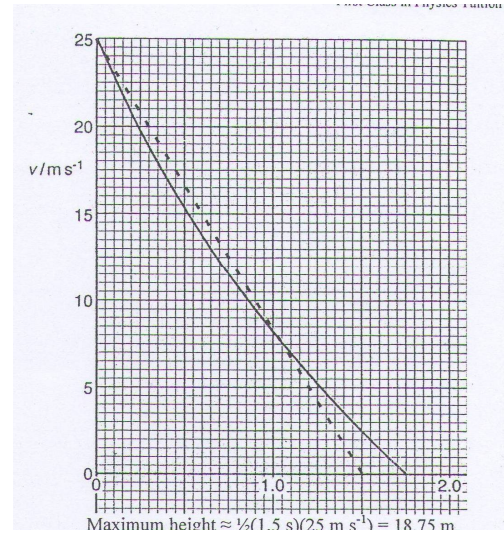
- 6 (a)** The gradient of the graph is not constant, indicating the acceleration of the ball changes with speed. From Newton's second law of motion, it can be deduced that the resultant force also changes with speed. Since the resultant force comprises the ball's weight and air resistance, but weight is constant, it may be deduced that the air resistance varies with speed.
- (b)** When the ball is at rest at $t = 1.75$ s, there is no air resistance hence the resultant force acting on the ball is its weight alone. The acceleration of the ball at this instant is the acceleration of free fall, which can be found from the gradient of the graph at $t = 1.75$ s.

**Many wrong answers did not specify the point that the tangent is to be drawn or some wrong answers specify that the tangent should be drawn at $t = 0$ s.*

- (c) (i) The maximum height is the area under the graph between $t = 0$ s and $t = 1.75$ s. This area can be approximated by finding the area under the dotted line in the figure shown, where the part of the graph which lies above the dotted line equalizes the part that lies below.

Maximum height

$$= \frac{1}{2}(1.5)(25) = 18.75 \text{ m}$$



**Inaccurate answers were obtained when the approximation was not done properly. This results in answers obtained to be out of the acceptable range of the question.*

- (ii) Energy loss due to air resistance = initial KE – gain in GPE

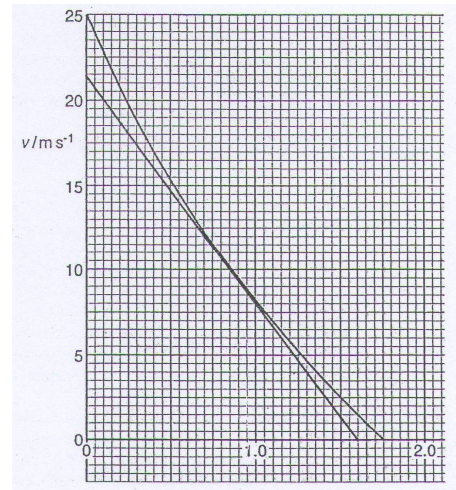
$$= \frac{1}{2}mv^2 - mgh$$

$$= \frac{1}{2}m(25)^2 - m(9.81)(18.75)$$

$$= 128.6m$$

$$\text{Ratio } E_T = \frac{128.6m}{\frac{1}{2}m(25)^2} = 0.41 \text{ since potential energy is zero.}$$

- (d) (i) Gradient of graph = $\frac{-21.5}{1.6} = -13 \text{ m s}^{-2}$

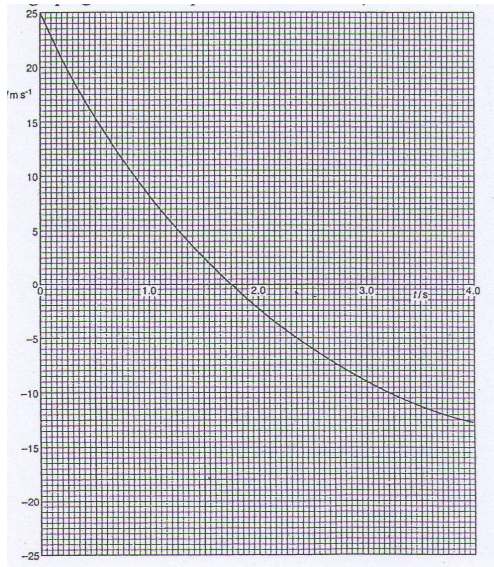


$$(ii) F = ma = (0.350)(-13) = -4.55 \text{ N}$$

$$-mg - R = -4.55 \text{ N}$$

$$R = 4.55 - mg = 4.55 - (0.35)(9.81) = 1.12 \text{ N}$$

(e)



**Areas bounded by the curve above and below the horizontal axis should be approximately equal.*

7. (a) (i) One volt (V) is the potential difference between two points in a circuit in which 1